

DNS Report

TryHackMe – DNS in Detail

Room covered: DNS in Detail

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Introduction

Define DNS in your own words.

Ans:- DNS (Domain Name System) is the internet's "phonebook". It translates human-friendly domain names such as `tryhackme.com` into the numerical IP addresses that computers use to find and talk to each other, so nobody has to memorise addresses like 104.26.10.229.

Explain why DNS knowledge is important in cybersecurity.

Ans:- Almost every connection on the internet starts with a DNS lookup, so DNS is both a **source of evidence** for defenders and a **target and tool** for attackers. A cybersecurity professional needs to understand it in order to:

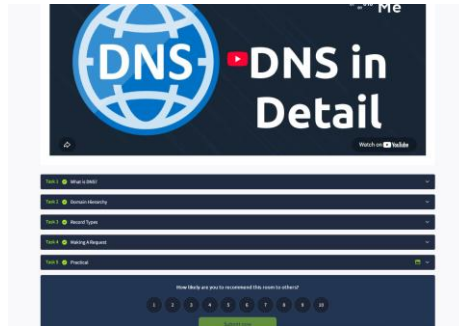
- **Spot malicious domains** – phishing sites often use look-alike (typosquatted) or newly registered domains.
- **Investigate incidents** – DNS logs show which domains a compromised machine contacted and when.
- **Detect DNS-based attacks** – DNS spoofing/cache poisoning redirects users to fake sites, and DNS tunnelling can hide data exfiltration or malware command-and-control traffic.
- **Secure email** – SPF, DKIM and DMARC policies are published as DNS TXT records.
- **Perform reconnaissance** – enumerating subdomains and records (with tools such as `nslookup`, `dig`) reveals an organisation's external footprint.
- **Troubleshoot networks** – many "the internet is down" problems are really DNS failures.

Overview of Assigned Room

One TryHackMe room was completed for this report: **DNS in Detail**. It explains how domain names are structured, what information DNS stores, how a lookup travels across the internet, and finishes with a hands-on practical using `nslookup`. The room contains five tasks: What is DNS, Domain Hierarchy, Record Types, Making a DNS Request, and Practical.

- **DNS:** system that maps domain names to IP addresses.
- **Domain hierarchy:** Root → TLD → Second-Level Domain → Subdomain.
- **DNS records:** A, AAAA, CNAME, MX and TXT – each stores a different type of information.
- **Recursive server:** resolver (usually run by the ISP) that performs the lookup on the client's behalf.
- **Authoritative server:** the server that holds the actual records for a domain.
- **TTL:** Time To Live – how long a DNS answer may be cached.
- **nslookup:** command-line tool used to query DNS servers.

TryHackMe – DNS in Detail room, completed



TryHackMe – DNS in Detail room overview (5 tasks)

Room 1 — DNS in Detail

Task 1: What is DNS?

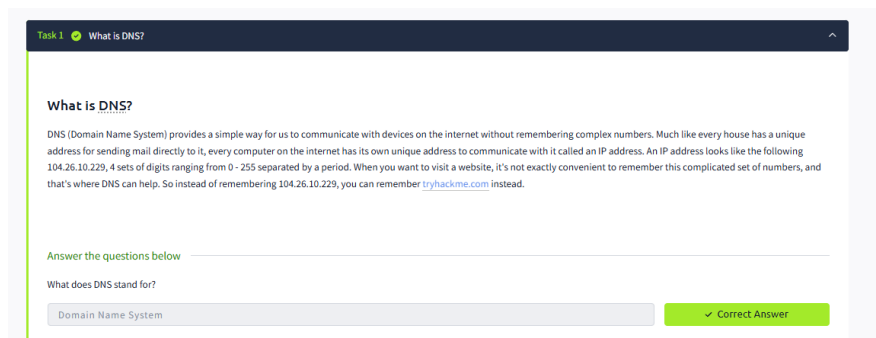
Objective

To understand what DNS is, the problem it solves, and why it is needed.

Information Covered

Computers communicate using **IP addresses**, but humans find names far easier to remember. DNS bridges this gap: instead of typing an address like 104.26.10.229, a user simply types **tryhackme.com**, and DNS resolves the name to the correct server's IP address in the background.

A useful comparison is the contacts list on a phone – you tap a person's name rather than dialling their number. DNS does the same job for websites and other internet services, and it also lets a service change its IP address without users ever noticing, because the name stays the same.



Task 1 room content

Tip: The question asks for the expansion of the acronym – it is given in the room's introduction.

Q1. What does DNS stand for?

Ans:- Domain Name System

Task 2: Domain Hierarchy

Objective

To learn how domain names are organised and the naming rules that apply at each level.

Information Covered

Domain names are organised as a **tree**, read from right to left – from the Root Domain at the top down to individual subdomains. Taking **jupiter.servers.tryhackme.com** as an example:

Level	Example	Description
Root Domain (.)	. (invisible dot)	Top of the hierarchy; branches into all TLDs.
Top-Level Domain (TLD)	.com	Rightmost part of a domain name.
Second-Level Domain (SLD)	tryhackme	Sits to the left of the TLD; the name that is registered.
Subdomain	servers, jupiter	Left of the SLD, separated by a period; can be chained.

Top-Level Domains (TLDs)

TLDs are split into two main types:

- **gTLD (generic TLD)** – indicates the purpose of a domain, e.g. **.com**, **.org**, **.edu**, **.gov**.
- **ccTLD (country-code TLD)** – tied to a country or territory, e.g. **.uk**, **.ca**, **.in**. A name such as **.co.uk** is therefore a ccTLD.

Second-Level Domains and Subdomains

Both share the same naming rules:

Rule	Detail
Maximum length	63 characters per label (SLD or subdomain).
Allowed characters	Letters a–z, digits 0–9 and hyphens.
Hyphen restrictions	Cannot start or end with a hyphen, and no consecutive hyphens.
Overall domain length	The complete domain name may be at most 253 characters.

Second-Level Domain

Taking [tryhackme.com](#) as an example, the .com part is the TLD, and tryhackme is the Second Level Domain. When registering a domain name, the second-level domain is limited to 63 characters + the TLD and can only use a-z 0-9 and hyphens (cannot start or end with hyphens or have consecutive hyphens).

Subdomain

A subdomain sits on the left-hand side of the Second-Level Domain using a period to separate it; for example, in the name [admin.tryhackme.com](#) the admin part is the subdomain. A subdomain name has the same creation restrictions as a Second-Level Domain, being limited to 63 characters and can only use a-z 0-9 and hyphens (cannot start or end with hyphens or have consecutive hyphens). You can use multiple subdomains split with periods to create longer names, such as [jupiter.servers.tryhackme.com](#). But the length must be kept to 253 characters or less. There is no limit to the number of subdomains you can create for your domain name.

Answer the questions below

What is the maximum length of a subdomain?

✓ Correct Answer 9

Which of the following characters cannot be used in a subdomain (3 b _ -)?

✓ Correct Answer

What is the maximum length of a domain name?

✓ Correct Answer

What type of TLD is .co.uk?

✓ Correct Answer

Task 2 room content – Second-Level Domain and Subdomain rules

Tip: All four answers are stated directly in the task's naming-rules text – match each question to the matching rule. No command is needed.

Q1. What is the maximum length of a subdomain?

Ans:- 63

Q2. Which character cannot be used in a subdomain (3, b, _ -)?

Ans:- _ (underscore)

Q3. What is the maximum length of a domain name?

Ans:- 253

Q4. What type of TLD is .co.uk?

Ans:- ccTLD

Task 3: DNS Record Types

Objective

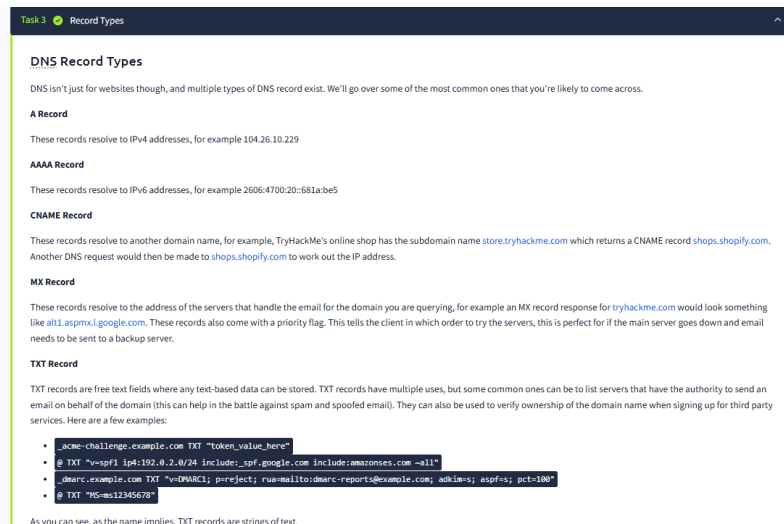
To understand the different kinds of information a DNS server can store about a domain.

Information Covered

DNS is not only about IP addresses. A domain's **records** hold several kinds of data, each with its own type:

Record	Full Name	Purpose	Example
A	Address	Maps a domain to an IPv4 address.	example.com → 104.26.10.229
AAAA	IPv6 Address	Maps a domain to an IPv6 address.	example.com → 2606:4700:20::681a:be5
CNAME	Canonical Name	Makes one domain an alias of another.	store.tryhackme.com → shops.shopify.com
MX	Mail Exchange	Names the mail server(s) and their priority.	tryhackme.com → mail server (lower priority number is tried first)
TXT	Text	Free-form text: domain verification, SPF, DMARC, tokens.	@ TXT "v=spf1 ..."

- **A / AAAA** – the records used for ordinary web browsing; AAAA exists because IPv6 addresses are longer than IPv4.
- **CNAME** – points a name at another name, so the final IP is resolved by a second lookup. Commonly used to point a subdomain at a third-party service.
- **MX** – tells other mail servers where to deliver email for the domain. Each entry has a priority; the server with the **lowest number** is tried first.
- **TXT** – used for email security policies (SPF/DMARC) and to prove ownership of a domain to third-party services.



Task 3 room content – A, AAAA, CNAME, MX and TXT records

Tip: Match each question to the “Purpose” column above – email routing is MX, and IPv6 addressing is AAAA.

Q1. What type of record would be used to advise where to send email?

Ans:- MX

Q2. What type of record handles IPv6 addresses?

Ans:- AAAA

Task 4: Making a DNS Request

Objective

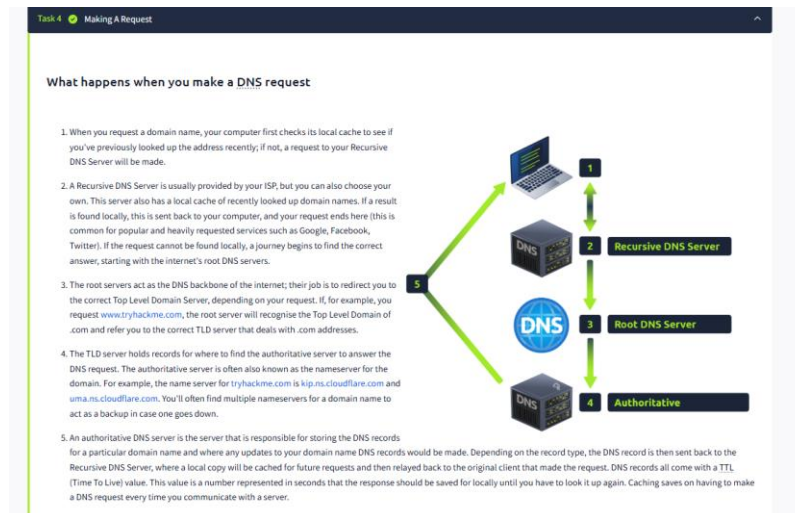
To follow the journey of a DNS query from the user’s computer to the server that finally answers it.

Information Covered

When a domain such as `tryhackme.com` is typed into a browser, the name is resolved through a chain of servers:

1. **Local DNS cache** – the device first checks whether it already knows the answer from a recent lookup.
2. **Recursive DNS server** – if not, the query goes to the recursive server (usually provided by the ISP). It checks its own cache and, if needed, queries other servers on the client’s behalf.
3. **Root DNS server** – the recursive server asks a root server, which directs it to the correct TLD server (e.g. for `.com`).
4. **TLD server** – points the recursive server to the authoritative server for the requested domain.
5. **Authoritative DNS server** – holds the actual records (A, AAAA, CNAME, MX, TXT) and returns the answer. The answer is passed back to the recursive server, then to the client, and is cached along the way.

Flow: Computer → Local cache → Recursive server → Root → TLD → Authoritative → answer returned to the computer.



Task 4 room content – DNS resolution flow

Server Types at a Glance

Server	Role
Recursive	Usually run by the ISP; does the lookup work for the client and caches results.
Root	Top of the lookup chain; refers the query to the correct TLD server.
TLD	Manages one TLD (e.g. <code>.com</code>) and points to the domain's authoritative server.
Authoritative	Stores the domain's records and gives the definitive answer.

TTL and Caching

Every DNS answer carries a **TTL (Time To Live)** value in seconds. It tells caches how long they may store the record before asking again. Caching makes DNS fast and reduces load on upstream servers; the trade-off is that a changed record can take up to its TTL to reach every user.

Answer the questions below

What field specifies how long a DNS record should be cached for?

TTL ✓ Correct Answer

What type of DNS Server is usually provided by your ISP?

recursive ✓ Correct Answer

What type of server holds all the records for a domain?

authoritative ✓ Correct Answer

Task 4 – answering the questions

Tip: Match each question to the steps above – the cache-duration field is TTL, the ISP-run server is Recursive, and the server storing all records is Authoritative.

Q1. What field specifies how long a DNS record should be cached for?

Ans:- TTL

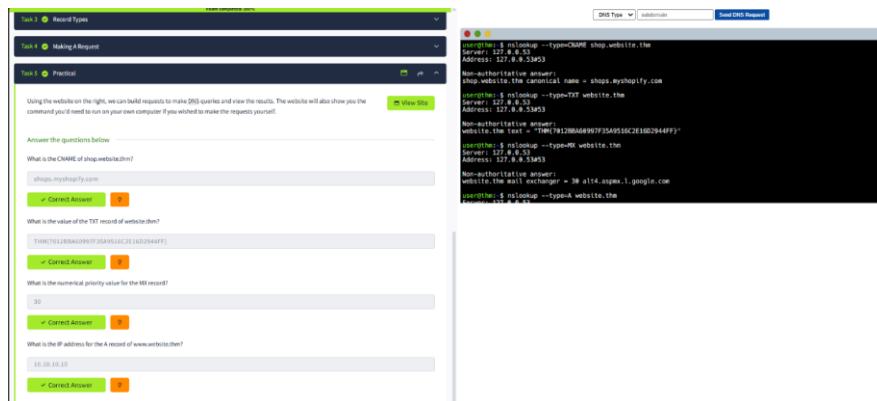
Q2. What type of DNS server is usually provided by your ISP?

Ans:- Recursive

Q3. What type of server holds all the records for a domain?**Ans:-** Authoritative**Task 5: Practical (nslookup)****Objective**To query real DNS records for **website.thm** using the **nslookup** command-line tool.**Information Covered****nslookup** sends queries to a DNS server and prints the reply. The record type is chosen with the **--type** option:

```
nslookup --type=<RECORD_TYPE> <domain>
```

After deploying the room's target, the terminal (or the on-screen "View Site" tool) is used to run each command and read the relevant line of the output.

*Task 5 room content – the on-screen tool and terminal for website.thm*

Tip: In the output, **Server: 127.0.0.53** is the local resolver the machine asked. “Non-authoritative answer” means the reply came from a resolver’s cache rather than directly from the domain’s authoritative server.

Step 1 — Find the CNAME of shop.website.thm

Query the subdomain and ask specifically for its CNAME record:

```
user@thm:~$ nslookup --type=CNAME shop.website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
shop.website.thm canonical name = shops.myshopify.com
```

The value after “canonical name =” is the alias target, which is the CNAME answer.

Q1. What is the CNAME of shop.website.thm?**Ans:-** shops.myshopify.com**Step 2 — Find the TXT record of website.thm**Change **--type** to TXT and query the root domain:

```
user@thm:~$ nslookup --type=TXT website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
website.thm text = "THM{7012BBA60997F35A9516C2E16D2944FF}"
```

The text in quotes after “text =” is the TXT record’s value – in this room it is the flag.

Q2. What is the value of the TXT record of website.thm?

Ans:- THM{7012BBA60997F35A9516C2E16D2944FF}

Step 3 — Find the MX priority for website.thm

Change **--type** to MX to find the mail server and its priority:

```
user@thm:~$ nslookup --type=MX website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
website.thm mail exchanger = 30 alt4.aspmx.l.google.com
```

The number immediately before the mail server name is the priority value (lower = tried first).

Q3. What is the numerical priority value for the MX record?

Ans:- 30

Step 4 — Find the A record (IP) for www.website.thm

Change **--type** to A and query the www subdomain to get its IPv4 address:

```
user@thm:~$ nslookup --type=A www.website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
Name: www.website.thm
Address: 10.10.10.10
```

The “Address:” line under the answer is the IPv4 address required.

Q4. What is the IP address for the A record of www.website.thm?

Ans:- 10.10.10.10

Quick Reference — All Commands and Results

Command	Result
nslookup --type=CNAME shop.website.thm	shops.myshopify.com
nslookup --type=TXT website.thm	THM{7012BBA60997F35A9516C2E16D2944FF}
nslookup --type=MX website.thm	alt4.aspmx.l.google.com, priority 30
nslookup --type=A www.website.thm	10.10.10.10

Task Summary: Putting It Together

The table below summarises the key ideas from all five tasks:

Term	Short Explanation
DNS	Translates domain names into IP addresses.
Domain hierarchy	Root → TLD → Second-Level Domain → Subdomain (max 63 characters per label, 253 in total).
Record types	A (IPv4), AAAA (IPv6), CNAME (alias), MX (mail), TXT (text/verification).
Resolution chain	Local cache → Recursive → Root → TLD → Authoritative.
TTL	How long a DNS answer may be cached.
nslookup	Command-line tool to query DNS records: <code>nslookup --type=<TYPE> <domain></code> .

Conclusion

DNS is a hierarchical system that converts human-friendly domain names into machine-usable IP addresses. It stores different kinds of data in records (A, AAAA, CNAME, MX and TXT) and answers queries through a resolution chain of recursive, root, TLD and authoritative servers. TTL-based caching keeps the process fast and efficient.

Completing this room provided both the theory (domain structure, record types and the lookup process) and practical experience running `nslookup` against different record types. Because DNS sits underneath almost all network activity, these are core skills for networking and for cybersecurity work such as investigating suspicious domains, analysing network traffic and securing email.